

Product Carbon Footprint Study Report

Product Name: light fuel oil

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: light fuel oil

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg light fuel oil

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg light fuel oil.

2. System Boundary

light fuel oil production in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of light fuel oil produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **0.785 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

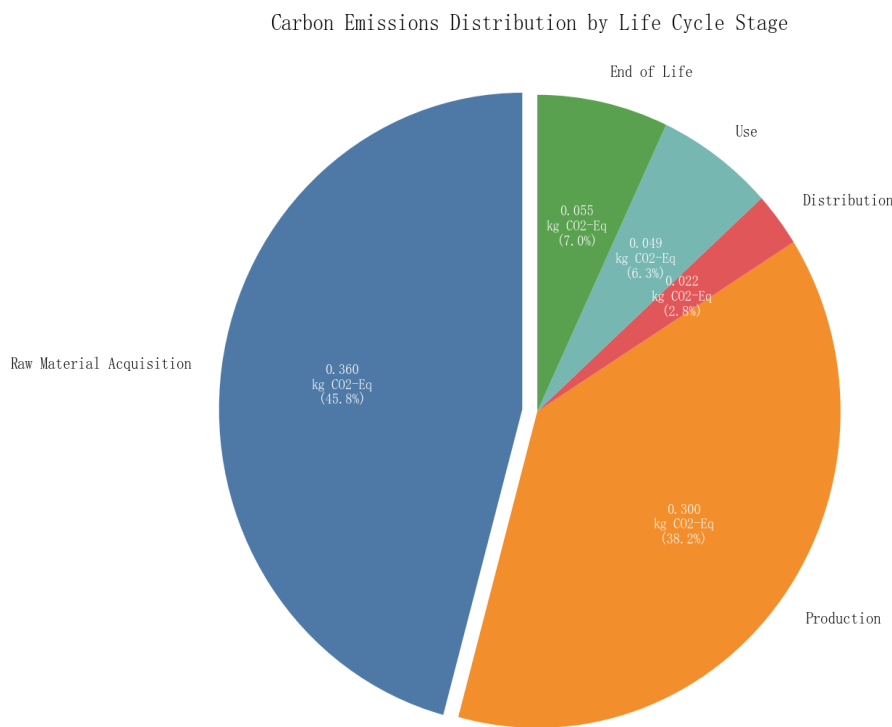


Table 1: Carbon Emissions by Life Cycle Stage

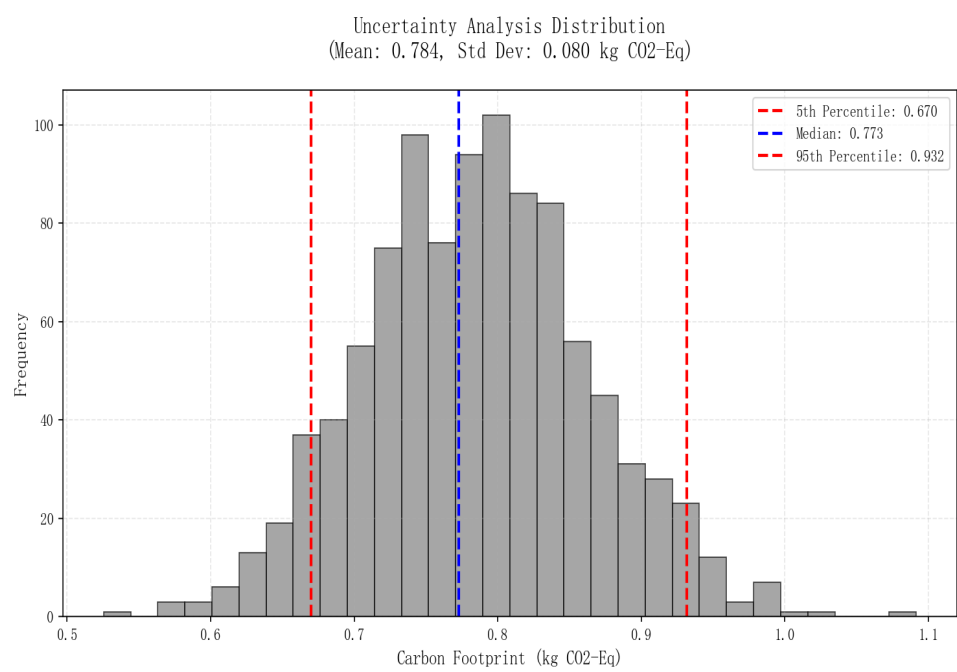
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.360	45.80
Production	0.300	38.18
Distribution	0.022	2.78

Use	0.049	6.26
End of Life	0.055	6.98
Total	0.785	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	0.784
Standard Deviation	0.080
Coefficient of Variation (CV)	10.18%
Median	0.773
5th Percentile	0.670
95th Percentile	0.932

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of light fuel oil is quantified at 0.785 kg CO₂-Eq, representing a moderate emission intensity typical for fossil-derived fuels when assessed on a per-unit basis. This value encompasses the complete cradle-to-grave perspective, highlighting the product's environmental burden across its entire value chain. The result positions light fuel oil as a contributor to greenhouse gas emissions that warrants targeted mitigation within energy and transport sectors.

Lifecycle contributions reveal a pronounced concentration in upstream stages, with raw_material accounting for 45.8% (0.360 kg CO₂-Eq) and production for 38.2% (0.300 kg CO₂-Eq), collectively responsible for over 84% of total emissions. Downstream phases remain marginal: distribution contributes 2.8%, use phase 6.3%, and end_of_life 7.0%. This distribution underscores extraction and refining as primary drivers, reflecting the inherent carbon intensity of fossil resource acquisition and processing operations.

Major emission sources are therefore identified as raw material extraction and production processes. These stages dominate due to energy-intensive activities such as drilling, refining, and associated fugitive emissions, emphasizing the need for interventions at the supply-chain origin rather than in logistics or consumption phases.

2. Statement of Assumptions and Limitations

Key assumptions in this study include the use of average European secondary datasets for fuel-specific processes, allocation of multi-output refinery burdens by economic value, and a functional unit defined per kilogram of light fuel oil delivered. The analysis presumes steady-state operational conditions without accounting for seasonal or regional variability in feedstock quality. Additionally, biogenic carbon is treated as neutral, and no credit is assigned for potential co-product utilization beyond the system boundary.

Limitations arise from reliance on generic database entries rather than site-specific primary measurements, which may introduce uncertainty of $\pm 15\text{-}20\%$ in emission factors for extraction and refining. Data quality is further constrained by temporal mismatches, as some inventory records predate recent efficiency improvements in upstream operations. Geographic representativeness is limited, potentially overlooking variations in regulatory standards or energy mixes across different production regions.

System boundary restrictions encompass a standard cradle-to-grave scope that excludes indirect land-use change, infrastructure construction, and end-user combustion equipment manufacturing. Downstream waste management is modeled only for packaging residues, omitting detailed treatment of combustion by-products or accidental spills, thereby narrowing the scope to core material and energy flows.

3. Suggestions for Improvement

- Prioritize low-carbon feedstock sourcing through certified suppliers or increased blending with biofuels to reduce the raw_material stage contribution by at least 20%.

- Implement process optimization and heat recovery systems in refining operations to lower production-phase emissions, targeting a 15% efficiency gain via best-available technologies.
- Establish supplier engagement programs for real-time energy monitoring and renewable electricity adoption at extraction sites to address the dominant upstream hotspots.
- Enhance data granularity by collecting primary site-specific measurements for the top two emission stages and updating inventory datasets annually.
- Explore circular economy pathways such as improved end-of-life recycling of associated materials to incrementally decrease the 7% end_of_life burden.